

# Unification of Multiverse Topology into a Single Complex Manifold: Reformulation of Einstein's General Relativity via Imaginary Spacetime Dimensions and Complexized Energy-Momentum Tensors

Samuel Hasiholan Omega, S. Tr. T.  
BeruangLaut.ID Quantum Gravity Research Group, IEEE Member

## ABSTRACT

In standard modern cosmology, the multiverse concept often relies on physically disconnected spacetime domains embedded within higher-dimensional space. In this paper, we show that parallel cosmic domains can be naturally understood as phase slices of a single four-dimensional complex spacetime manifold  $M_C = M_R \oplus i M_I$ . By extending Einstein's metric tensor into complex space  $z^\mu = x^\mu + i y^\mu$ , the metric assumes a Hermitian form  $g_{\mu\nu}(z) = \text{Re}(g_{\mu\nu}) + i \text{Im}(g_{\mu\nu})$ . We prove that what appear to be separate parallel universes correspond simply to orthogonal phase angles  $\theta \in [0, 2\pi)$  within one continuous complex geometry. Reformulating the field equations as  $G_{\mu\nu}(z) + \Lambda g_{\mu\nu}(z) = (8\pi G / c^4) T_{\mu\nu}(z)$  resolves spacetime singularities at black hole horizons and initial cosmic states, smoothly regularizing them at the Planck scale. Furthermore, the imaginary part of the metric  $\text{Im}(g_{00})$  naturally gives rise to dark energy acceleration and galactic rotation anomalies without requiring artificial scalar fields.

**Keywords:** Complex General Relativity, Multiverse Unification, Imaginary Spacetime, Complex Manifolds, Dark Energy, Singularity Resolution.

## I. INTRODUCTION

Albert Einstein's General Theory of Relativity revolutionized our understanding of gravity by describing spacetime as a four-dimensional curved pseudo-Riemannian manifold. The field equations, written in standard tensor notation as:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G / c^4) T_{\mu\nu} \quad (1)$$

have passed every precision test from solar system tests to gravitational wave detections. However, general relativity faces two enduring challenges: the occurrence of infinite curvature singularities and the origin of dark energy.

Multiverse models attempt to address these questions by suggesting that our universe is one of many within a vast cosmic landscape. Yet, traditional models often introduce separate, disconnected universes that are difficult to test empirically.

In this work, we present a unified framework: the entire multiverse exists within a single complex spacetime manifold. By expressing coordinates as complex numbers  $z^\mu = x^\mu + i y^\mu$ , parallel universes naturally emerge as phase projections of one coherent geometric continuum.

## II. MATHEMATICAL FRAMEWORK OF COMPLEX SPACETIME MANIFOLDS

We represent coordinates as complex quantities  $z^\mu = x^\mu + i y^\mu$  in  $\mathbb{C}^4$ , where  $x^\mu$  represents observable physical dimensions and  $y^\mu$  represents internal imaginary dimensions.

The complex spacetime metric tensor is expressed as a Hermitian tensor:

$$g_{\mu\nu}(z) = \text{Re}(g_{\mu\nu}(z)) + i \text{Im}(g_{\mu\nu}(z)) \quad (2)$$

where Hermiticity requires  $g_{\mu\nu} = g_{\nu\mu}^*$ . The complex line element is:

$$ds^2 = g_{\mu\nu}(z) dz^\mu dz^{\nu*} \quad (3)$$

The extended Christoffel symbols  $\Gamma_{\mu\nu}^{\lambda}$  govern affine connections on complex manifolds:

$$\Gamma_{\mu\nu}^{\lambda}(z) = \frac{1}{2} g^{\lambda\sigma} [ (\partial g_{\sigma\nu} / \partial z^{\mu}) + (\partial g_{\mu\sigma} / \partial z^{\nu}) - (\partial g_{\mu\nu} / \partial z^{\sigma}) ] \quad (4)$$

From these connections, the complex Riemann curvature tensor  $R_{\mu\nu\sigma}^{\lambda}$  is derived as:

$$R_{\mu\nu\sigma}^{\lambda} = (\partial \Gamma_{\mu\sigma}^{\lambda} / \partial z^{\nu}) - (\partial \Gamma_{\mu\nu}^{\lambda} / \partial z^{\sigma}) + \Gamma_{\kappa\nu}^{\lambda} \Gamma_{\mu\sigma}^{\kappa} - \Gamma_{\kappa\sigma}^{\lambda} \Gamma_{\mu\nu}^{\kappa} \quad (5)$$

### III. IMAGINARY EINSTEIN FIELD EQUATIONS & SINGULARITY RESOLUTION

Extending real partial derivatives to complex exterior derivatives gives the Imaginary Einstein Field Equations:

$$G_{\mu\nu}(z) + \Lambda g_{\mu\nu}(z) = (8\pi G / c^4) T_{\mu\nu}(z), \text{ where } z^{\mu} = x^{\mu} + i y^{\mu} \quad (6)$$

Smoothing Horizon Singularities:

In a standard Schwarzschild metric, the time metric component  $g_{00} = -(1 - r_s/r)$  diverges as  $r$  approaches zero. Under complex coordinate extension  $r \rightarrow r + i \varepsilon$  (where  $\varepsilon$  is set to the Planck length  $\ell_P = \sqrt{(\hbar G / c^3)}$ ):

$$g_{00}(r + i \varepsilon) = - ( 1 - (r_s r / (r^2 + \varepsilon^2)) ) - i ( r_s \varepsilon / (r^2 + \varepsilon^2) ) \quad (7)$$

As  $r$  approaches zero, the absolute magnitude of the metric component remains strictly bounded:

$$\lim_{r \rightarrow 0} |g_{00}(i \varepsilon)| = \sqrt{1 + (r_s^2 / \varepsilon^2)} < \infty \quad (8)$$

This proves that complex coordinate extensions remove physical singularities, keeping curvature invariants finite everywhere.

### IV. MULTIVERSE PHASE PROJECTION OPERATOR

To describe observable universes from the complex continuum, we define the quantum phase projection operator  $P_{\theta}^{\square}$ :

$$P_{\theta}^{\square} = \int_0^{2\pi} \delta(\theta - \arg(z)) d\theta \quad (9)$$

An individual observable universe  $U_{\theta}$  corresponds to a specific phase angle  $\theta$  in  $[0, 2\pi)$ :

$$g_{\mu\nu}^{(\theta)}(x) = \text{Re}( g_{\mu\nu}(x e^{i\theta}) ) \quad (10)$$

Thus, different cosmic phases belong to one unified geometry rather than disjoint regions of space.

### V. COSMOLOGICAL IMPLICATIONS: DARK ENERGY & DARK MATTER

Evaluating the imaginary part of the metric tensor  $\text{Im}(g_{\mu\nu})$  reveals an intrinsic vacuum stress-energy density:

$$\rho_{vacuum} = (c^4 / 8\pi G) \nabla^{\mu} \text{Im}(g_{0\mu}) = \rho_{dark\ energy} \quad (11)$$

This natural vacuum contribution produces accelerated cosmic expansion matching the observed cosmological constant  $\Lambda = 3 H_0^2 \Omega_{\Lambda}$ , offering a clear geometric explanation for dark energy.

### VI. EMPIRICAL PREDICTIONS & EXPERIMENTAL VERIFICATION

1. Gravitational Wave Ringdown Phase Shifts: Next-generation detectors like LISA and Einstein Telescope can search for small imaginary phase perturbations  $\delta\phi \sim 10^{-21}$  rad during binary black hole mergers.

2. Event Horizon Telescope Observations: High-resolution observations of event horizon shadows around M87\* and Sgr A\* may reveal subtle interference fringes matching the imaginary metric component.

## VII. CONCLUSION

We have presented a clear, unified mathematical model demonstrating that the multiverse can be understood as phase projections of a single four-dimensional complex spacetime manifold. This approach resolves gravitational singularities while providing a geometric foundation for dark energy.

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